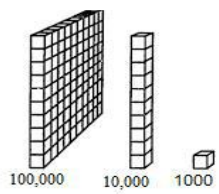
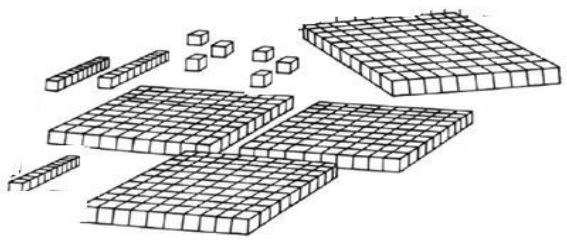
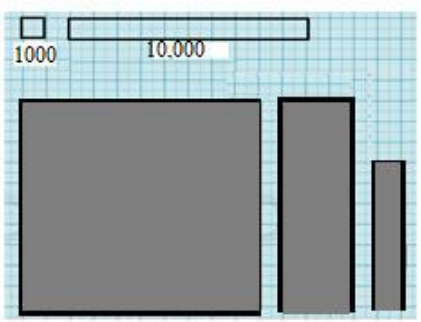
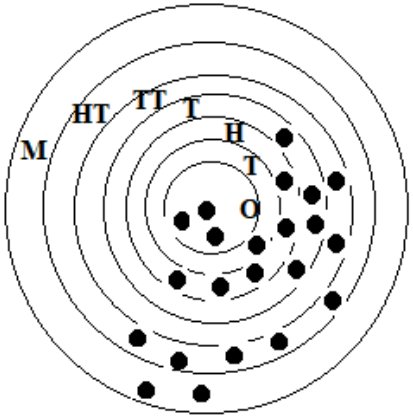
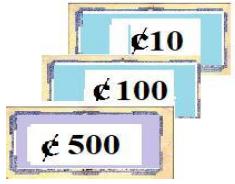
BASIC 5

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.1 **Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1000,000**	**B5.1.1.1.1 Model number quantities up to 1,000,000 using graph sheets and multi-base block** E.g. 1. Ask pupils to model number quantities up to 1000,000 using graph sheets or multi-base materials. For instance, with multi-base block, a cube = 1000 unit; a rod = 10,000; a flat = 100,000 and a block = 1,000,000; learners model 436,000 with the appropriate materials E.g. 2. Ask pupils to model the number 137,000 shading graph sheet square as shown below, where 1cm×1cm square represents 1000 units .   	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.1 **Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1000,000. CONT'D**	E.g.3.Give learners teacher-make token currency notes [¢10, ¢100& ¢500 notes] on different coloured-paper and ask them to work out how many will be required to model given amounts up to ¢10,000; e.g. ¢23,480 **B5.1.1.1.2 Read and write numbers in figures and in words up to 1000,000** E.g. 1 Play the place value number wheel game: Use the million number wheel to generate 7-digit numbers and represent the number generated on a place value frame. Throw a number of pebbles (or stones) onto the number wheel, identify the number of pebbles landing in each place value ring, and state the number generated in the wheel (i.e. 2 landed in the ten-thousands' ring making the number twenty-thousand or 20,000)  Place value number wheel Place value chart Million | Hundred-thousands | Ten-thousands | Thousand | Hundreds | Tens | Ones 2 | 5 | 2 | 4 | 5 | 1 | 3 E.g. 2. Ask pupils to write given numbers on the expanded form of the number : 2,524,513=1000,000× 2 +100,000× 5 + 10,000 × 2+1000 × 4+ 100× 5 + 10× 1+ 1× 3 = 2,000,000 + 500,000 + 20,000 + 4,000 + 500 + 10 + 3  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Cultural Identity

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<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr>
<td>B5.1.1.1

Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1000,000. CONT'D</td>
<td>
B5.1.1.1.3 Identify numbers in different positions around a given number in a number chart

E.g. 1 Display a number chart in multiples of 1,500 between 10,000 and 1000,000 and lead learners identify numbers in different positions around a given number. Put learners in pairs and give each group a number grid and have them identify numbers in different positions around a chosen number
<table>
<tr><td>10,000</td><td>11,500</td><td>12,000</td><td>13,500</td><td>15,000</td><td>16,500</td></tr>
<tr><td>20,000</td><td>21,500</td><td>22,000</td><td>23,500</td><td>24,000</td><td>25,500</td></tr>
<tr><td>30,000</td><td>31,500</td><td>33,000</td><td>34,500</td><td>36,000</td><td>37,500</td></tr>
<tr><td>40,000</td><td>41,500</td><td>43,000</td><td>44,500</td><td>46,000</td><td>47,500</td></tr>
<tr><td>50,000</td><td>51,500</td><td>53,000</td><td>54,500</td><td>56,000</td><td>57,500</td></tr>
<tr><td>60,000</td><td>61,500</td><td>63,000</td><td>64,500</td><td>66,000</td><td>67,500</td></tr>
</table>
B5.1.1.1.4 Compare and order whole numbers up to 100,000 and represent the comparison using ">, <, or ="

E.g. 1 Learners identify numbers which are 10,000 more or 10,000 less than a given six-digit number. E.g. 122,400 is 1,000 less than133,400 Learners use words such as: "equal to" "greater than" "less than" and later use symbols such as "<", "=", ">" to compare numbers up to 10,000 taking into consideration the place value of each digit in the given number. E.g. $251200 = 251{,}200;\ 132{,}734 > 132{,}635$

E.g. 2 Learners work together in their groups to order a given set of numbers in ascending or descending order verbally and in writing. E.g. $1020, 1025,\ 2673, 2873,$
</td>
<td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Cultural Identity</td>
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<tr><td>B5.1.1.1
Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1000,000. CONT'D</td><td>E.g. 3 Give two numbers between 10,000 and 100,000 to each group and encourage learners to say as many things as possible about the two numbers. For instance 1022345 and 1022534; 1022345 is less than (smaller than) than1022534 or 1022534 is bigger than (greater than) 1022345, or 1022345 is almost 200 less than 1022534 etc

B5.1.1.1.5 Round (off, up, down) whole numbers up to 100,000 to the nearest ten thousands, thousands, hundreds and tens

E.g. 1 Learners round off numbers to the nearest 10,000, 1000 and 100. For instance; 129,500 is approximately 130,000 and 19100 as approximately 19,000

E.g. 2 Learners round up and round down to estimate sums and differences

NB: To round down learners consider the smaller of the two approximated numbers, while round up considers the bigger of the two numbers. The table below may bring the meaning of these two concepts clearer

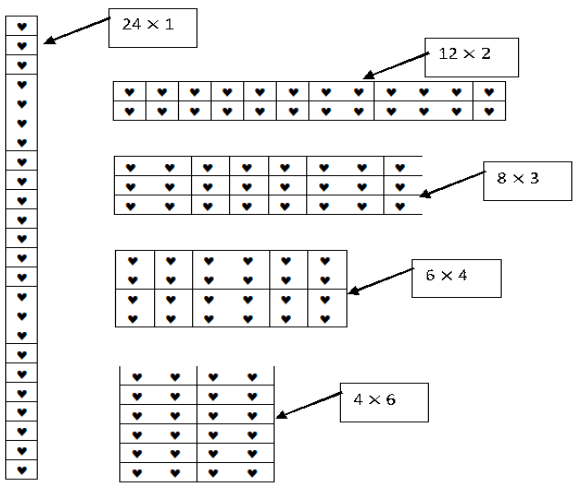
<table><tr><th>214765</th><th>Round up</th><th>Round down</th><th>Round off</th></tr><tr><td>to the nearest ten</td><td>214770</td><td>214760</td><td>214770</td></tr><tr><td>to the nearest hundred</td><td>214800</td><td>214700</td><td>214800</td></tr><tr><td>to the nearest thousand</td><td>215000</td><td>214000</td><td>215000</td></tr></table></td><td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Cultural Identity</td></tr>
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.1 **Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1,000,000.**	**B5.1.1.1.6 Skip count forwards and backwards in 500s 1000s etc. up to and from 100,000** E.g. 1 Put learners into convenient groups. A learner mention a number and another makes skip count in 500s or 1000s to include the fifth count. For instance. Learner 1: Shout out "15290"*skip counting up in 500s* Learner 2: 15790, 16290, 17290, 17790, 18290, etc. Learner 3; Shout out "31285"*skip counting up in 1000s* Learner 4; 32285, 33285, 34285, 35285, 36285….etc. E.g. 2 Skip count forwards and backwards by 50s and between 1000 and 10000 that are multiples of 50s and 100s but make an error or leave out a number. Challenge learners to identify or correct error.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Cultural Identity
B5.1.1.2 Demonstrate understanding of Roman Numerals up to C (i.e. 100)	**B5.1.1.2.1 Recognize Roman Numerals system up to C (i.e. 100)** E.g. 1 Display roman numeral charts (1-100) arranged in sequential order and lead learners to identify the numerals. Learners identify the main characters of the roman numerals used to build the table up to 30 i.e. I, II, III, IV, V, X, L and C Call out a numeral and have learners point at it from the chart E.g. 2 Have learners match the Roman numerals to the Hindu-Arabic numerals for instance I = 1; V= 5: IX= 9, XV= 15, XXX = 30, XL = 40, LVI = 56, XCIX = 99. Mention some numerals randomly and have learners point at it on the chart. **B5.1.1.2.2 Count and convert Hindu Arabic numbers to Roman numerals up to 100 (C) and vice versa** E.g. 1 Learners read the chart sequentially forwards and backwards, vertically (up and down), zig-zag, diagonally and randomly without pointing to the numbers. Invite 2-3 learners to read to the class. Call a Roman numeral and ask learners to write E.g. 2 Give learners a numeral in the Hindu Arabic system and have learners convert to roman numeration and vice versa for instance XXIV = 24, LX = 60, XCIV = 94, ETC.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Cultural Identity

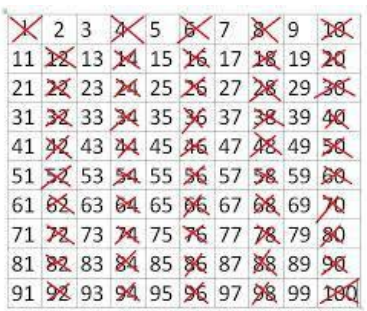
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.3 **Demonstrate an understanding of factors, multiples of numbers including composite, even, odd and prime numbers from 1 to 100**	**B5.1.1.3.1 Identify the factors of whole numbers 1 - 100** E.g. 1 Ask learners to use different arrangements of arrays of objects to find factors of whole numbers 1 – 100. For instance the factors of 24 can be obtained from the following arrays of objects. E.g. 2 Ask learner to collect from the arrays, pairs of number as the factors of the given number, 24. Hence the set of factors of $24 = \{1, 2, 3, 4, 6, 8, 12, 24\}$  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Cultural Identity

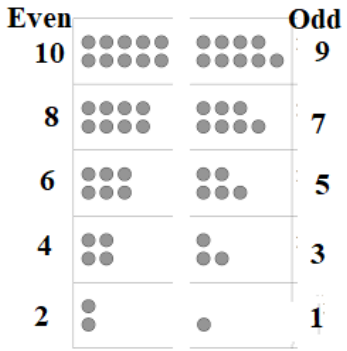
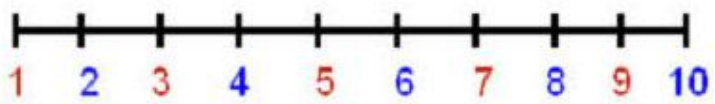
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.3 **Demonstrate an understanding of factors, multiples of numbers including composite, even, odd and prime numbers from 1 to 100**	**B5.1.1.3.2 Generate and identify prime numbers and composite numbers between 1 and 100** E.g. 1 Ask learners to use different arrangements of arrays of objects (or the factorization method)to find factors of the first 30 whole numbers and use their results to complete the table below. Have learners identify prime numbers; i.e. number with only two factors, 1 and the number itself; other are composite. E.g. 2 In convenient groups, have learners use the "Sieve of Eratosthenes" to identify prime numbers between 1 and 100, i.e. dividing through by whole numbers starting from 2; numbers that cannot be divided by other numbers except the number itself are prime numbers.	

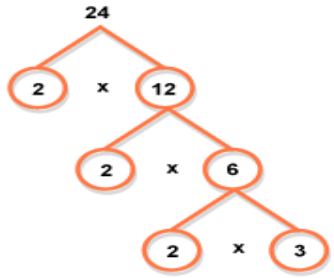
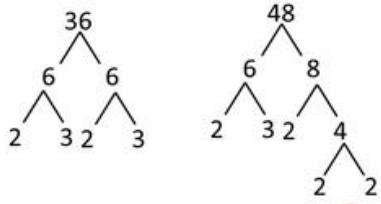
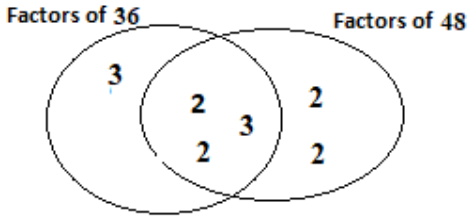
Numbers	Factors	Number of factors
1	1	1
2	1, 2	2
3	1, 3	2
4	1, 2, 4	3
5	1, 5	2
.		
.		
.		
30	1,2, 3, 5, 6, 10, 15, 30	



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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.1 Demonstrate an understanding of quantities and place value for multi-digit numerals up to 100,000 CONT'D	**B5.1.1.3.3 Identify even and odd numbers between 1 and 100 as number that can be arrayed in twos array and those which cannot** E.g. 1 Ask learners to use arrangements of twos arrays of objects to find even and odd numbers between 1 and 100  E.g. 2 Ask learner to collect from the arrays, pairs to skip count in twos starting from 1 to generate odd numbers, and skip count in twos starting from 2 to generate even numbers  	Learners develop Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.3 Demonstrate an understanding of factors, multiples of numbers including composite, even, odd and prime numbers from 1 to 100 CONT'D	**B5.1.1.3.4 Determine the highest common factor of any 2 or 3 numbers by prime factorisation** E.g. 1. Have learners use factor tree method to determine prime factors of any given whole number. For example the prime factors of 24 For instance, from the figure 24 = 2×2×2×3 so 2 and 3 are the prime factors of 24. Ask learners to list the factors of two or more given whole numbers using the factor tree; for 36 and 48 we have 36= 2×3×2×3 =2×2×3×3 and 48= 2×3×2×2×2 = 2×2×2×2×3.  E.g. 2 Learners use the prime factorization to determine the HCF by underlining the common factors in each product 36=**2×2×3**×3 48= 2×2×**2×2×3** → which is 2×2×3=12;  E.g. 3 Learners use the prime factorization by inspection to determine the LCM by underlining the largest number of factors in each product 36=2×2×**3×3** 48= **2×2×2×2**×3 → which is 2×2×2×2×3×3=144  	Learners develop: Problem solving skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.1.3 Demonstrate an understanding of factors, multiples of numbers including composite, even, odd and prime numbers from 1 to 100	E.g. 4 Learners place factors in a Venn diagrams to find the HCF and LCM of 36 and 48. i.e. the HCF is product of factors in both circles →2×2×3=12; and the LCM is product of factors in the diagram →3×2×2×3×2×2 =144 E.g. 5 Learners use the prime factorization to determine the LCM and HCF of three numbers using (i) prime factorization using table and dividing through by prime factors. Example 1, find the LCM and HCF these sets of numbers: 12, 20 and 30. (see method in figure), i.e. HCF = 2, and LCM = 2×2×3×5 = 60.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision

*2	12	20	30
2	6	10	15
3	3	5	15
5	1	5	5
	1	1	1

The prime factor which can divide all the numbers is starred. The product of all the starred prime factors is the HCF. In this case only one is starred, and so the HCF is 2.

The product of each prime factor which divided any of the numbers is the LCM

B5.1.1.3.5. Recognize relationship between factors and multiples of whole numbers from 1 to 100

E.g. 1 Investigate even and odd numbers. How do you know a number is even or odd?

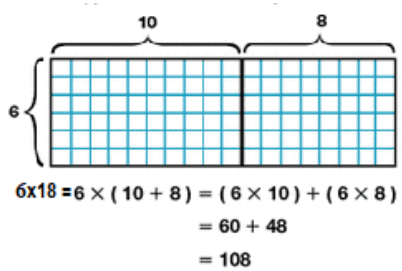
E.g. 2 Investigate numbers that are multiples of 4 and 6. How do you know a number is a multiple of 4? a multiple of 6? (This is also known as the divisibility test).

E.g. 3 Investigate perfect numbers, that is, numbers whose factors add up to the number; for instance 6 has factors 1, 2, 3 and 6. The sum of factors other than 6 is 1+2+3 =6, and hence 6 is a perfect number. How many more perfect number can we find in the first 100 whole numbers?

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.1 **Describe and apply mental mathematics strategies and number properties to determine answers for basic multiplication facts to 81 and related division facts**	**B5.1.2.1.1 Apply mental mathematics strategies and number properties, such as skip counting from a known fact, using doubling or halving, using patterns in the 9sfacts, using repeated doubling or halving, to determine answers for basic multiplication facts to 81 and related division facts** E.g. 1 Describe the mental mathematics strategies used to determine a given basic fact, such as – skip count up by one or two groups from a known fact (e.g., if 5 × 7 = 35, then 6 × 7 is equal to 35 + 7 and 7 × 7 is equal to 35 + 7 + 7) – skip count down by one or two groups from a known fact (e.g., if 8 × 8 = 64, then 7 × 8 is equal to 64 – 8 and 6 × 8 is equal to 64 – 8 – 8) – doubling (e.g., for 8 × 3 think 4 × 3 = 12, and 8 × 3 = 12 + 12) – patterns when multiplying by 9 (e.g., for 9 × 6, think 10 × 6 = 60, and 60 – 6 = 54; for 7 × 9, think 7 × 10 = 70, and 70 – 7 = 63) – repeated doubling (e.g., if 2 × 6 is equal to 12, then 4 × 6 is equal to 24 and 8 × 6 is equal to 48) – repeated halving (e.g., for 60 ÷ 4, think 60 ÷ 2 = 30 and 30 ÷ 2 = 15) – relating division to multiplication (e.g., for 64 ÷ 8, think 8 × … = 64) E.g. 2 Recall multiplication facts to 81and related division facts	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision

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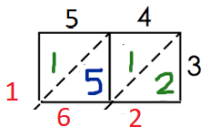
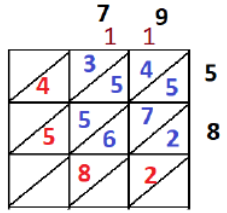
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.1 **Describe and apply mental mathematics strategies and number properties to determine answers for basic multiplication facts to 81 and related division facts**	**B5.1.2.1.2 Apply mental mathematics strategies for multiplication, such as annexing then adding zero halving and doubling using the distributive property** E.g. 1 Determine the products when one factor is a multiple of 10, 100, or 1000 by annexing zero or adding zeros (e.g., for 3×200 think of 3×2 and then add two zeros) E.g. 2 Apply halving and doubling when determining a given product (e.g., 32×5 is the same as 16×10) E.g. 3 Apply the distributive property to determine a given product involving multiplying factors that can be written in the expanded form (e.g., 6×18 = (6×10) + (6×8) = 60+48 = 108. Or apply distributive property to determine a given product involving multiplying factors that are close to multiples of 10 (e.g., 29×7 = (30×7) – (1×7) = 203. E.g. 4 Explain the property for determining the answer when multiplying numbers by  i. one ii. zero.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.2 **Demonstrate an understanding of multiplication of a 2 or 3-digit number by a 2 or 3-digit number.**	**B5.1.2.2.1. Multiply multi digit numbers by 2-digit numbers efficiently** E.g. 1. Multiplication of whole numbers using the "expand and box" method (partial decomposition method). Lead learners to multiply a 3-digit number by a 1-digit number as shown below, for example $448 \times 2 = ?$ $448 \times 2 = (400 \times 2) + (40 \times 2) + (8 \times 2) = 800 + 80 + 16$ $= 800 + 80 + (10 + 6) = 800 + 90 + 6 = 896$ E.g. 2. Multiplication of whole numbers using the Column or vertical method; i.e. 25×32 E.g. 3. Multiplication of whole numbers using the Distributive property; i.e. 25×32 $25 \times (32) = 25(30 + 2)$ $= 25(30) + 25(2)$ $= 750 + 50$ $= \mathbf{800}$ 25 x32 50 +750 **800** 25 25 x30 x 2 750 50 = **800**	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision

×	400	40	8
2	800	80	16

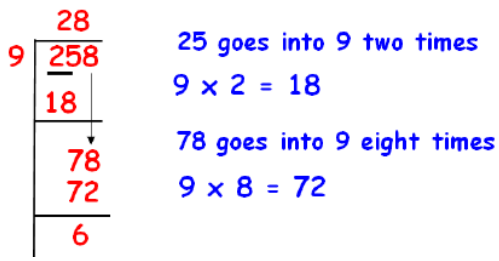
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.2 **Demonstrate an understanding of multiplication of a 2 or 3-digit number by a 2 or 3-digit number.** **CONT'D**	E.g. 4. Multiplication of whole numbers using the lattice method. Have learners draw a 2 by 1 rectangular box for solving 54 × 3.   E.g. 5. Have learners draw a 2 by 2 rectangular box for solving 79×58.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision

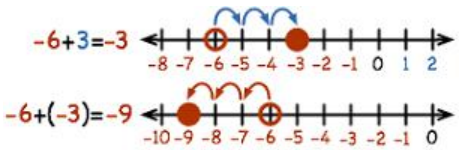
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.3 **Recall basic division fact up to 100**	**B5. 1.2.3.1 Determine basic division fact up to 81** E.g. 1. Investigate numbers that are multiples of 3,4, 6, 8 and 9. How do you know a number is divisible by 3, 4, 6, 9? (This is also known as the divisibility test). E.g. 2. Draw a 6 by 6 multiplication chart and use it as a game board to play the 3-in-a-line game with a pair of dice or playing cards. Players take turns in throwing a one dice and mark (or cover) a number that can be divided by the results (i.e. a quotient). The winner is the one who obtains three numbers in a line.   		5	6	7	8	9	10
---	---	---	---	---	---	---
5	10	15	20	25	30	35
6	12	18	24	30	36	42
7	14	21	28	35	42	49
8	16	24	32	40	48	56
9	18	27	36	45	54	63
10	20	30	40	50	60	70		Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.4 **Demonstrate an understanding of division of a 2 or 3-digit number by a 1 or 2-digit number**	**B5.1.2.4.1 Divide 3-digit numbers by 1-digit number efficiently** E.g. 1. Division as repeated subtraction (using the long division method). Explain division as a way of repeatedly subtracting a divisor number from a given dividend until there is none left and then determining the number of times the divisor is taken from the dividend  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision
B5.1.2.5 **Translate word problems into mathematical sentences and solve**	**B5.1.2.5.1. Solve multi step word problems involving the four basic operations** E.g. 1. Learners model mathematical statements from a given word problem involving addition and multiplication and solve using the strategies learnt E.g. 2. Learners model mathematical statements from a given word problem involving division and subtraction and solve using the strategies learnt E.g. 3. Learners role play a given word problem involving addition and multiplication and solve E.g. 4. Learners role play a given word problem involving subtraction and division and solve E.g. 5. Learners model mathematical statements from a given word problem involving division and multiplication and solve using the strategies learnt. The relationship between operations and the use of calculator and spreadsheet to assess the reasonableness of answers should be stressed.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.2.6 **Demonstrate understanding of integers**	**B5.1.2.6.1 Solve simple addition and subtraction problems involving integers** E.g. 1. Use number line to help learners to do the following types (addition) (1) 9 + -4 = _____ (2) -8 + 4 = _____ (3) -3 + -5 = _____ (4) 1 + -3 = _____ (5) -6 + 5 = _____ (6) 6 + -2 = _____ (7) -6 + 8 = _____ (8) -2 + 9 = _____  E.g. 2 Use number line to help learners to do the following types (subtraction) (9) -5 – 1 = ____ (10) -2 – 1 = ____ (11) 8 – 7= ____ (12) 2 – 6 = _____ (13) -1 – 7 = ____ (14) -5 – 7= ____ (15) -8 – 8= ____ (16) 4 – 6 = _____ E.g. 3. Solve word problems; e.g. (i) Some number added to 5 is equal to -11. Find the number. (ii) Ben and Cam are diving. Ben is 15.8 meters below the surface of the water. Cam is 4.2 meters above Ben What is Cam's position relative to the surface of the water?	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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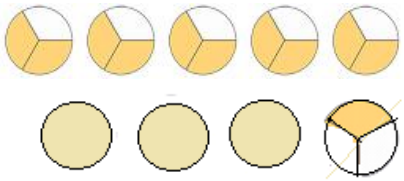
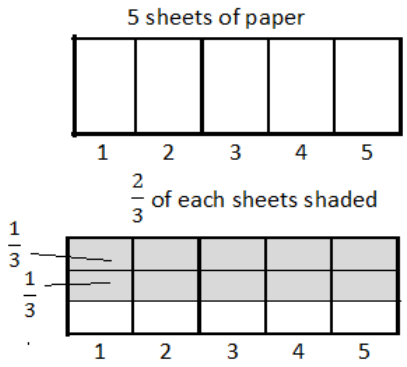
<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td>B5.1.3.1
Demonstrate understanding of strategies for comparing, adding, subtracting and multiplying fractions</td><td>B5.1.3.1.1 Determine equivalent fractions of given fractions
E.g. 1 To compare, add or subtract the fractions, $\frac{3}{4}, \frac{5}{6}$ and $\frac{7}{10}$, we need find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions. Use the prime factorising strategy to determine the LCD of 4, 6, and 20. Use the LCD work out the equivalent fractions.
$\frac{3}{4} = \frac{?}{36}; \frac{5}{6} = \frac{??}{36}$; and $\frac{7}{10} = \frac{???}{36}$.
[To obtain the numerators, determine how many times the denominator goes into the LCD and multiply this by the numerator to obtain a new numerator.]
24 = 2 x 2 x 2 x 3 so 2 and 3 are prime factors of 24
<table><tr><td>2</td><td>4</td><td>6</td><td>10</td></tr><tr><td>2</td><td>2</td><td>3</td><td>5</td></tr><tr><td>3</td><td>1</td><td>3</td><td>5</td></tr><tr><td>5</td><td>1</td><td>1</td><td>5</td></tr><tr><td></td><td>1</td><td>1</td><td>1</td></tr></table>The product of each prime factor which divided any of the numbers is the LCM</td><td rowspan="2">Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
<tr><td>B5.1.3.1
Demonstrate understanding of strategies for comparing, adding, subtracting and multiplying fractions</td><td>B5.1.3.1.2 Compare and order fractions
E.g. 1 Find the Lowest Common Denominator (LCD) of the fractions and use this to express their equivalent fractions and use them for the comparison; e.g. which is larger, $\frac{5}{6}$ and $\frac{3}{4}$ or $\frac{2}{3}$?
E.g. 2 Change the fractions to percentages or decimals and use them for the comparison.
i.e. $\frac{5}{6} = 0.83$, $\frac{3}{4} = 0.75$, and $\frac{2}{3} = 0.66$, and compare.
LCD for $\frac{3}{4}, \frac{2}{3}$ and $\frac{5}{6}$ is 12;
hence $\frac{3}{4} = \frac{9}{12}; \frac{2}{3} = \frac{8}{12}; \frac{5}{6} = \frac{10}{12}$
$\therefore$ the order is $\frac{5}{6}, \frac{3}{4}, \frac{2}{3}$</td></tr>
</table>

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.3.1 **Demonstrate understanding of strategies for comparing, adding, subtracting and multiplying fractions CONT'D**	**B5.1.3.1.3 Use the concept of equivalent fractions in changing fractions to the same denominator for addition and subtraction of like fractions (one denominator being a multiple of the other)** E.g. 1 To add or subtract the fractions, $\frac{1}{3}$ and $\frac{2}{6}$, we need find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions. The LCD is 6 and the equivalent fractions are $\frac{2}{6}$ and $\frac{2}{6}$; their sum is $\frac{2}{6} + \frac{2}{6}$ is $\frac{2+2}{6}$, which is, $= \frac{4}{6} = \frac{2}{3}$ (i.e. its simplest form) E.g. 2 To add or subtract the fractions, $\frac{2}{3}$ and $\frac{2}{5}$, we need find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions. The LCD is 15 and the equivalent fractions are $\frac{10}{15}$ and $\frac{6}{15}$; their difference is $\frac{10}{15} - \frac{6}{15} = \frac{10-6}{15} = \frac{4}{15}$ **B5.1.3.1.4 Use the concept of equivalent fractions for addition and subtraction of fractions greater than one (improper or mixed fractions)** E.g. 1 To add like mixed fractions that are larger than 1, i.e. $2\frac{1}{3}$ and $3\frac{2}{3}$, we write down the sum of the whole numbers and add the fractions; i.e. $2\frac{1}{3} + 3\frac{2}{3} = 5 + \frac{1}{3} + \frac{2}{3}$, $= 5\frac{1+2}{3} = 5\frac{3}{3} = 6$. E.g. 2 To subtract like-fractions that are larger than 1, i.e. $2\frac{1}{3}$ and $3\frac{2}{3}$, we change the mixed fractions into improper fractions and subtract; i.e. $3\frac{2}{3} - 2\frac{1}{3} + = \frac{11}{3} - \frac{7}{3} + = \frac{11-7}{3} = \frac{4}{3}. = 1\frac{1}{3}$ E.g. 3 To add or subtract improper fractions with different denominators, ($2\frac{1}{3}$ and $3\frac{2}{5}$) we need find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions. The LCD is 15 and the equivalent fractions are $2\frac{5}{15}$ and $3\frac{6}{15}$; their sum is $2\frac{1}{3} + 3\frac{2}{5} = 2\frac{5}{15} + 3\frac{6}{15} = 5\frac{5+6}{15}$	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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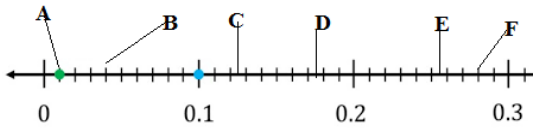
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.3.1 **Demonstrate understanding of strategies for comparing, adding, subtracting and multiplying fractions CONT'D**	which is $5\frac{11}{15}$; and difference $3\frac{2}{5} - 2\frac{1}{3} = 1\frac{6-5}{15} = 1\frac{1}{15}$ **B5.1.3.1.5 Use models to explain the result of multiplying a whole number by a fraction** E.g. 1 Multiplying a whole number by a fraction, e.g. $5 \times \frac{2}{3}$ or finding five two-thirds means $\frac{2}{3} + \frac{2}{3} + \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = \frac{10}{3} = 3\frac{1}{3}$  E.g. 2 To multiply a whole number by a mixed fraction (e.g. $3 \times 2\frac{2}{3}$) one can multiply the whole number by the whole number and then whole number by the fraction and add the products or change the mixed fraction to improper fraction and multiply; i.e. $3 \times 2\frac{2}{3} = (3\times2) + 3\times\frac{2}{3}) = 6 + \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = 6\frac{6}{3} = 8$ or $3 \times 2\frac{2}{3} = 2\frac{2}{3} + 2\frac{2}{3} + 2\frac{2}{3} = 6\frac{2+2+2}{3} = 6\frac{6}{3} = 8$ E.g. 3 To multiply a whole number by a fraction (e.g. $3 \times 2\frac{2}{3}$) first change all into common fractions, then multiply the numerators separately and multiply the denominators separately and simplify; i.e. $3 \times 2\frac{2}{3} = \frac{3}{1} \times \frac{8}{3} = \frac{3\times8}{1\times3} = \frac{24}{3} = 8$ **B5.1.3.1.6 Use models to explain the result of multiplying a fraction by whole number** E.g. 1 Multiplying a fraction by a whole number the 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.3.1 **Demonstrate understanding of strategies for comparing, adding, subtracting and multiplying fractions CONT'D**	multiplication is interpreted as "of"; e.g. $\frac{2}{3} \times 5$ means shade $\frac{2}{3}\ of\ 5$; i.e. finding two-thirds of each of five objects; i.e. $\frac{2}{3} \times 5$ can be illustrated by shading $\frac{2}{3}\ of\ 5$ sheets of paper, which leads to the shading of 10 thirds, $\frac{2}{3} \times 5 = \frac{2}{3}\ of\ 5 = 10(\frac{1}{3}) = \frac{10}{3} = 3\frac{1}{3}$ E.g. 2 To multiply a mixed fraction by a whole number (e.g. $4\frac{2}{5} \times 5$) first change all into common fractions, then multiply the numerators separately and multiply the denominators separately and simplify; i.e. $4\frac{4}{5} \times 5 = \frac{24}{5} \times \frac{5}{1} = \frac{120}{5} = \frac{24}{1} = 24$	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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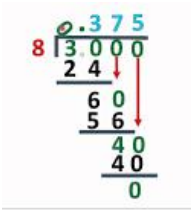
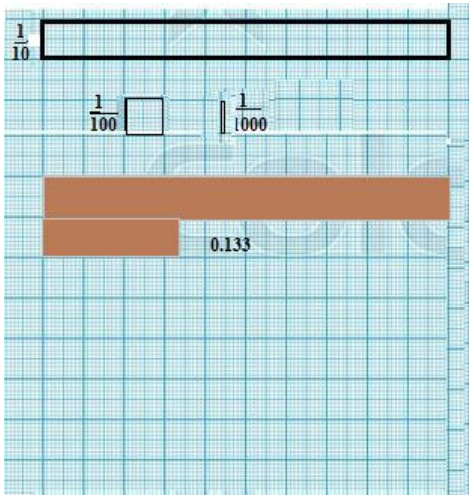
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<tr>
<td>B5.1.4.1
Demonstrate understanding of decimals fractions and strategies for comparing, adding and subtracting decimal fraction</td>
<td>
B5.1.4.1.1 Describe and represent decimals (up to the thousandths) concretely, pictorially, and symbolically and relate them to fractions

E.g. 1 Use several pictorial representations or number line to introduce tenths and thousandths and ask pupils to identify the fractions (i.e. A, B, C, D, E, & F)


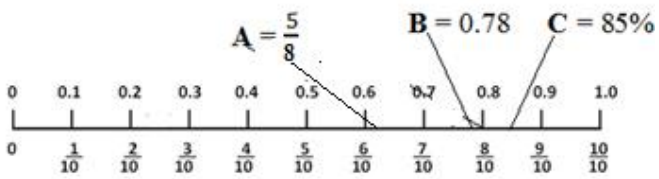
E.g. 2 Use the tenth to thousandth place value chart to illustrate how to write the fraction as a base ten number with the introduction of the decimal point "·" after ones place in the place value chart
<table>
<tr><th colspan="6">Tenth to Thousandth Place Value Chart</th></tr>
<tr><th>Fraction</th><th>1s</th><th>·</th><th>$\frac{1}{10}$s</th><th>$\frac{1}{100}$s</th><th>$\frac{1}{1000}$s</th></tr>
<tr><td>A</td><td>$\frac{1}{4}=\frac{25}{100}$</td><td>0</td><td>·</td><td>5</td><td></td><td></td></tr>
<tr><td>B</td><td>$\frac{1}{8}=\frac{125}{1000}$</td><td>0</td><td>·</td><td></td><td></td><td></td></tr>
<tr><td>C</td><td>$2\frac{1}{40}$</td><td>0</td><td>·</td><td></td><td></td><td></td></tr>
<tr><td>D</td><td>$1\frac{3}{8}$</td><td>0</td><td>·</td><td></td><td></td><td></td></tr>
</table>
</td>
<td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
</tr>
</table>

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.4.1 **Demonstrate understanding of decimals fractions and strategies for comparing, adding and subtracting decimal fraction CONT'D**	**B5.1.2.1.1. Multiply multi-digit numbers by 2-digit numbers efficiently** E.g. 3 Give pupils several common fractions (including improper fractions) to convert into hundredths and thousandths and write their decimal names; e.g. $\frac{3}{8}$  E.g. 4 Ask pupils to use graph sheets to model the decimal number 0.133 by shading the graph sheet squares as shown in the figure, where 2cm×20cm square represents $\frac{1}{10}$ units, 2cm×2cm square represents $\frac{1}{100}$ units, and 1mm×10mm square represents $\frac{1}{1000}$ units, 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.4.1 **Demonstrate understanding of decimals fractions and strategies for comparing, adding and subtracting decimal fraction CONT'D**	**B5.1.4.1.2 Compare and order decimal fractions and a mixture of common and decimal fractions (up to the thousandths) using the symbols <, >, or =** E.g. 1 To compare and order a mixture of common, decimal and percent fractions, express them in one form (i.e. either common, decimal or percent); e.g. to order 0.758, $\frac{5}{8}$ and 73%; we have $\rightarrow\frac{5}{8} = \frac{625}{1000}$ = 62.5%, $0.758 = \frac{758}{1000}$ = 75.8%, and $73\% = \frac{73}{100}$ = 0.73, Hence the order from least to the largest is $\frac{5}{8}$, 73% and 0.758 E.g. 2 To compare and order a mixture of common, decimal and percent fractions you can locate the fractions on the number and order them  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.4.1 **Demonstrate understanding of decimals fractions and strategies for comparing, adding and subtracting decimal fraction**	**B5.1. 4.1.3 Round decimals to the nearest tenth and hundredth** E.g. 1 Explain the rule for of rounding decimals, which is similar to that of rounding whole numbers. That is, rounding to the nearest tenth means that the rounded figure has one digit after the decimal mark. Rounding to the nearest hundredth means that the rounded figure has two digits after the decimal mark Rounding 85674.875 rounded to 10 ▶ 85670 100 ▶ 85700 1 decimal or $\frac{1}{10}$ place ▶ 85674.9 1 decimal or $\frac{1}{100}$ places ▶ 85674.88 E.g. 2 Use a table with several decimals fractions and explain the rule for of rounding decimals. Give pupils a table of decimal fractions to round to the nearest tenths or hundredths. E.g. 3 Ask students to change fractions to decimal writing their results to the nearest tenths or hundredths.	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

	Fraction	Round to Nearest	Result
A	0.38	$\frac{1}{10}$	0.4
B	4.085	$\frac{1}{10}$	
C	56.584	$\frac{1}{10}$	
D	18.096	$\frac{1}{100}$	
E	30.084	$\frac{1}{100}$	

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.4.1 **Demonstrate understanding of decimals fractions and strategies for comparing, adding and subtracting decimal fraction CONT'D**	**B5.1.4.1.4 Use models to explain the result of addition and subtraction of decimals (up to the thousandths)** E.g. 1 To add 0.645 and 0.39, means sum $\frac{645}{1000} + \frac{39}{100}$ which is $\frac{645+390}{1000} = \frac{1035}{1000}$ = 1.035. That is, since one is hundredths and other thousandths (i.e. one has 2 decimal places and the other 3 decimal places) we can line up the decimal points to align the values and add as whole numbers $\begin{array}{r} 0.645 \\ +\,0.390 \\ \hline 1.035 \end{array}$ and the place E.g. 2 To subtract 0.395 from 0.6 and, one number is in tenths and other is in thousandths (i.e. one has 1 decimal place and the other 3 decimal places) it is important to line up the decimal points to align the place values and whole numbers. $\begin{array}{r} 0.6 \\ -\,0.396 \\ \hline 0.204 \end{array}$ add as **B5.1.4.1.5 Use models to explain the result of multiplying a decimal (up to the thousandths) by a whole number** E.g. 1 Review multiplying a fraction by a whole number. e.g. $\frac{2}{5} \times 10$ means shade $2 \times (\frac{1}{5}\ of\ 10)$; i.e. 2×(2) = 4; E.g. 2 To multiply a decimal fraction by a whole number, change the decimal number to common fraction and multiply e.g. 0.4×10 means $\left(\frac{4}{10}\ of\ 10\right) = 4.0$; Or multiply the whole numbers 4 and 10, and place the decimal point at one decimal place, since 0.4 has one decimal place E.g. 3 To multiply a decimal fraction (up to thousandths) by a whole number, first ignore the decimal point and multiply the whole numbers, check the number of decimal places and place the decimal point	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.5.1 **Demonstrate understanding of percentage of a given number**	**B5.1.5.1.1 Determine the percentage of a given quantity (limit to 2-digit whole number) and vice versa** E.g. 1 Review multiplying a fraction by a whole number. e.g. $\frac{2}{5} \times 10$ means shade $2 \times (\frac{1}{5}\ of\ 10)$; i.e. 2×(2) = 4; E.g. 2 To multiply a percent fraction by a whole number, change the percentage to common fraction and multiply and simplify e.g. $40\% \times 10$ means $\left(\frac{40}{100}\ of\ 10\right) = 4$; E.g. 3 Ask learners to find given percentage of given quantities; e.g. What is 20% of 45? What is 20% of 45? $20\%\ \text{of}\ 45 = 20\% \times 45$ $= \frac{20}{100} \times \frac{45}{1}$ $= 9$	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.5.1 **Demonstrate understanding of percentage of a given number CONT'D**	**B5.1.5.1.2 Determine the benchmark percentages from their common fractions and use these to estimate percentages of quantities** E.g. 1 Use pictorial representations and chart to display common or benchmarks percentages and ask pupils to determine these from their equivalent common fractions. E.g. 2 Give learners practice through drills and games to learn the equivalences of the benchmark fractions E.g. 3 Ask pupils to use the benchmarks for estimating percentages of given quantities. E.g. for "what is 60% of 45?" using the nearest benchmark fraction (i.e. 50%) the learner will know the expected result is close to 30;the learner can use benchmark fractions to determine the result mentally by finding which can easily multiply 45, and in this case $\frac{1}{5}$ to give 9. Since $\frac{1}{5}$ is 20%, 60% required will be 3 times 9 which is 27. E.g. . Ask pupils to use the benchmarks for estimating and determining the results of finding percentages of given and then verify by working; that is, changing the percentage common fraction and multiplying by the quantity	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

Percent	10%	20%	25%	$33\frac{1}{3}\%$	50%	$66\frac{2}{3}\%$
Fraction	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$

What is 60% of 45?

$$60\% \text{ of } 45 = 60\% \times 45$$
$$= \frac{\cancel{60}^{3}}{\cancel{100}_{5}} \times \frac{\cancel{45}^{9}}{1}$$
$$= 27$$

then the quantities to

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.1.5.1 **Demonstrate understanding of percentage of a given number CONT'D**	**B5.1.5.1.3 Identify and describe percent from real-life contexts and solve problems using percent.** E.g. 1 Discuss the contexts below in which fractions are used in real life and provide materials for pupils to act them out – In shops – discounts, reduction to clear – 50%, etc. – In exams expressing marks as percentages – Phone battery power used etc.   E.g. 2 Drawing circle graphs that represent that represent various percentages of halves, fourths and eighths. The graph shows the ages of pupils in Primary 4. If there are 40 pupils in the class, ask questions for pupils to interpret the graph 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development

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Strand 2. ALGEBRA

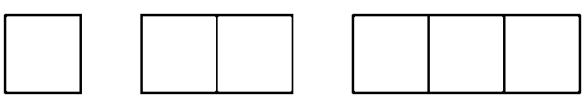
Sub-strand 1: Patterns and Relationships

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr>
<td>B5.2.1.1
Determine the pattern rule to make predictions about subsequent elements.</td>
<td>B5.2.1.1.1 Extend a given pattern with and without concrete materials, and explain how each element differs from the preceding one
E.g. 1 Skip count in multiples of 20, 50, 500, etc. is one way to introduce patterns.
E.g. 2 Provide students with a thousand number chart or number line with missing numbers and help use the strategy of skip counting to find missing numbers

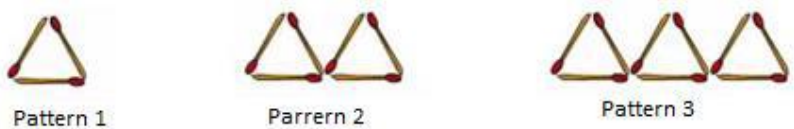
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<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>470</td><td>480</td><td></td><td></td></tr>
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<tr><td>610</td><td></td><td></td><td>640</td><td></td><td></td><td></td><td></td><td>690</td><td></td></tr>
<tr><td>710</td><td></td><td></td><td></td><td>750</td><td></td><td></td><td>780</td><td></td><td></td></tr>
<tr><td></td><td></td><td>830</td><td></td><td></td><td></td><td>870</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>930</td><td>940</td><td>950</td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
</td>
<td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
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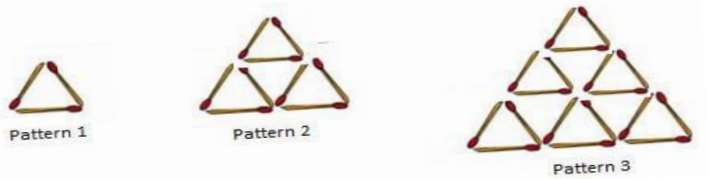
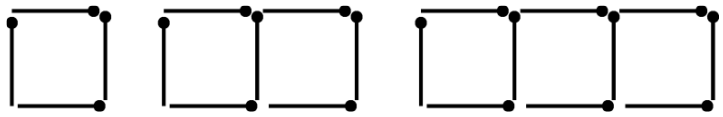
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.2.1.1 **Determine the pattern rule to make predictions about subsequent elements CONT'D**	**B5.2.1.1.2 Describe, orally or in writing, a given pattern, using mathematical language, such as one more, one less, five more, one more than or less than twice, etc.** E.g. 1 Lines are drawn to intersect two lines in each diagram and the number of points of intersection are counted to form a sequence. Ask learners to tabulate your results for the next four terms in the sequence and complete the table Find the rule for *l*ines  **Number of lines** | 1 | 2 | 3 | 4 | 5 | 6 | · | · | **Intersections** | 2 | 4 | | | | | | | Guide learners to describe the relationship as "*intersection is one less number of lines"* E.g. 2 Consider a design on the wall is being made using square tiles of size 10cm by 10cm  Ask learners to find the number of joints made by the tiles if one, two, three, four, etc. of the squares tiles are used, and use it to complete the following table and describe the relationships. **Number of tiles** | 1 | 2 | 3 | 4 | 5 | | | | **Number of joints** | 0 | 1 | | | | | | |	Learners develop: Problem Solving kills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.2.1.1 **Determine the pattern rule to make predictions about subsequent elements CONT'D**	**B5.2.1.1.3 Predict subsequent elements in a given pattern** E.g. 1 Ask students to respond to find the next two terms in a pattern and state the rule for the pattern. • What are the next two terms and what is the pattern? 2, 4, 6, 8, 10, 12, ..., ..., +2 +2 +2 +2 +2 +2 +2 Two must be ADDED to the previous TERM 3, 6, 9, 12, 15, 18, ..., ..., +3 +3 +3 +3 +3 +3 +3 Three must be ADDED to the previous TERM E.g. 2 Ask learners to complete the table below for the number of match sticks used in this geometric pattern. Guide learners to describe the relationship.  Pattern 1 Parrern 2 Pattern 3 E.g. 3 Ask learners to write questions based on number or geometric patterns for their friends to find answers to; e.g. how many match sticks will be used for the 9th geometric pattern?	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

Number of triangles	1	2	3	4	5				*n*
Number of match sticks	3	6							

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.2.1.1 **Determine the pattern rule to make predictions about subsequent elements CONT'D**	**B5.2.1.1.4 Represent a given pattern visually to verify predictions** E.g. 1 Ask learners to examine the pattern in the table below which shows the number of match sticks used in making a pattern of triangles.  **Pattern Number (sticks in side)** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 **Number of match sticks** | 3 | 9 | 18 | | | | | | Ask learners to use match sticks to make the patterns of triangles and complete the table. Guide learners to describe the relationship E.g. 2 Ask learners to examine the pattern in the table below which shows the number of match sticks used in making a pattern of squares Ask learners to use match sticks to make the patterns of triangles and complete the table. Guide learners to describe the relationship  **Pattern Number (sticks in side)** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 **Number of Pattern 1 triangles** | 1 | 4 | 9 | | | | | | E.g. 3 Ask learners to write questions based on number or geometric patterns for their friends to find answers to; e.g. how many match sticks will be used for the 9th pattern of squares? How many match sticks will be used for the 8th pattern of squares?	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td>B5.2.1.1
Determine the pattern rule to make predictions about subsequent elements CONT'D</td><td>
B5.2.1.1.5 Solve a given problem by using a pattern rule to determine subsequent elements
E.g. 1. Ask learners to describe the pattern rule for each example and determine the next three elements of each:
(i) 1, 10, 7, 70, 67, 670, ...
(ii) 10, 12, 16, 22, 30 ...
(iii) 50, 48, 47, 45, 44 ...

E.g. 2. Ask learners to describe the pattern rule for each of the patterns and determine the next three elements:
(i) .25, .5, .75, ___, ___, ___,
(ii) 2.50, 5, 7.50, ___, ___, ___,
(iii) 64, 32, 16, , ___, ___, ___,
(iv) 900, 450, 225, , ___, ___, ___,

B5.2.1.1.6 Determine and explain why a given number is or is not the next element in a pattern
E.g. 1 Ask students to respond to the following prompt: Shika filled bags with marbles. She placed two marbles in the first bag, four marbles in the second bag, six marbles in the third bag, eight marbles in the fourth bag and twelve in the fifth bag.
Her friend Ayerko noticed an error in the pattern. Can you identify and describe the error?
<table>
<tr><td>Bags</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr>
<tr><td>Marbles</td><td>2</td><td>4</td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
B5.2.1.1.7 Write a rule in words and in algebra to represent a given pattern
E.g. 1. E.g. 1. Ask learners to state the rules in words and in algebra to represent a given increasing linear patterns.
</td><td>Learners develop:

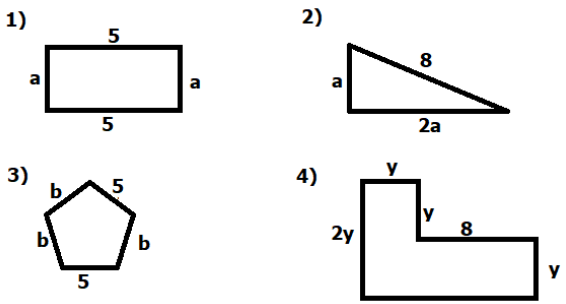
Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
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<tr>
<td>B5.2.1.1
Determine the pattern rule to make predictions about subsequent elements CONT'D</td>
<td>
<table>
<tr><th>Term/Input</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>Rule for n in words</th><th>Rule for n in Algebra</th></tr>
<tr><td>Result/Output A</td><td>7</td><td>14</td><td>21</td><td></td><td></td><td></td><td>7 times n</td><td>7n</td></tr>
<tr><td>Result/Output B</td><td>0</td><td>4</td><td>8</td><td>12</td><td></td><td></td><td>4 times one less n</td><td>4(n-1)</td></tr>
<tr><td>Result/Output C</td><td>4</td><td>7</td><td>10</td><td></td><td></td><td></td><td>1 more than 3 times n</td><td>1 + 3n</td></tr>
<tr><td>Result/Output D</td><td>5</td><td>9</td><td>13</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Result/Output E</td><td>5</td><td>11</td><td>17</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Result/Output F</td><td>0</td><td>6</td><td>12</td><td>18</td><td></td><td></td><td></td><td></td></tr>
</table>
B5.2.1.1.8 Describe the relationship in a given table or chart, using a mathematical expression
E.g.1 This table shows the pattern of cost of boxed lunches for students on a field trip. Ask learners to: (i) explain the pattern of how the cost of lunches changes as more students go on the trip; (ii) use the pattern to determine how many students went on the trip if the cost of lunches is GHC90.
<table>
<tr><td>Number of students</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>?</td></tr>
<tr><td>Cost of lunch in cedis</td><td>3</td><td>6</td><td>9</td><td>12</td><td>15</td><td>90</td></tr>
</table>
E.g. Ask learners to write questions based on the pattern for their friends to find answers.
</td>
<td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision</td>
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.2.2.1 **Demonstrate understanding of algebraic expressions**	**B5.2.2.1.1 Demonstrate understanding of algebraic expressions as mathematical phrases that can contain letters which represent ordinary numbers and operators (like add, subtract, multiply, and divide).** E.g. 1. Write algebraic expressions for word problems: 1) Sum of 8 and s 2) 8 times the sum of c and 7 3) Take away 4 from m 4) Subtract 4 from 7 times g E.g. 2. Write algebraic expressions for the perimeter of the following shapes: 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
	E.g. 2. Simplify basic algebraic expressions by grouping like terms. E.g. 1) m + 5 m 2) b + (–3b) + b 3) –7 n + 6 n 4) 9 w – 4 – 10 w 5) 8 w + 5 w 6) – m + 9 – 5 m E.g. 3. Substitute a value for an unknown into an expression and correctly calculate the answer. E.g. Find the values of these algebraic expressions using the values given in the following: 1) z + 7d use z = 7 and d = 4 2) 4 - 5s + 3b use s = 6 and b = 2 3) 8k + d use k = 2 and d = 3 4) -7 - 2b + 6 - 3r use b = 3 and r = 4 5) 7(5f - 3n) - 8 use n = 3 and f = 7 6) -5d - k 7 _ use k = 14 and d = 5	

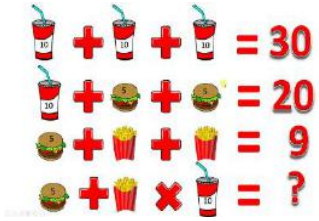
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<tr>
<td>B5.2.3.1
Solve problems involving single-variable, one-step equations with whole number coefficients</td>
<td>
B5.2.3.1.1 Express a given problem as an equation where the unknown is represented by a letter to variable.
E.g.1. Ask learners solve word problems; e.g..
i. The cost of two pens is twenty ¢15. If one costs ¢5.50, what is the cost the other pen?
ii. The product of two numbers is 120. If one of the numbers is 24, what is the other number?

E.g.2. Ask learners to read Ama's problem below, complete the table and solve the problem
<table>
<tr><td>Days (d)</td><td>Days</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td></td></tr>
<tr><td>Height (h)</td><td>Height</td><td>5cm</td><td>7cm</td><td></td><td></td><td></td><td></td></tr>
</table>
Ama is tracking the progress of her plant's growth. Today the plant is 5cm tall. The plant grows 2cm a day
(i) Write a mathematical sentence that represents the height of the plant after d days;
(ii) What will the height of the plant be after 20 days?
(iii) How many days will the height take to reach 75cm?
</td>
<td>Learners develop:

Problem Solving skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
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</table>

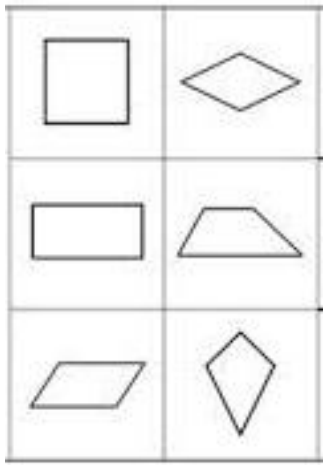
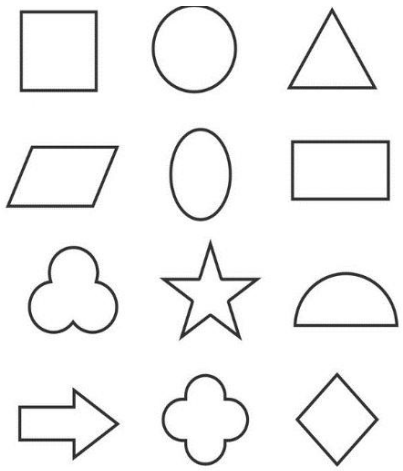
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.2 **Perform a single transformation (translation and reflection) of a 2D shape**	**B5.2.3.1.2 Identify the unknown in a problem; represent the problem with an equation; and solve the problem concretely, pictorially or symbolically.** E.g.1 Learners use concrete materials, such as blocks or counters and the balance scales, to find the value of p in the following equations. If necessary, model the use of guess and test as one strategy. By observing patterns in their results, students become more systematic in the guesses they make 1. 3 + p =11 2. 14 - p = 8 3. p - 9 = 16 4. 3p = 12 5. p + 5 = 17 6. 24 = p + 9 **7.** 25 = 35 - p **B5.2.3.1.3Create a problem for a given equation** E.g.1 Ask students to describe two different stories that the equation 5 + k = 9 could represent E.g.2. Ask learners to solve puzzle in the figure, by solving the equations in each line 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning Look for Patterns and Relationships

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Strand 3: GEOMETRY AND MEASUREMENT

Sub-Strand 1: Lines and Shapes

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.1.1 **Identify and sort quadrilaterals, including rectangles, squares, trapezoids, parallelograms, and rhombuses, according to their attributes**	**B5.3.1.1.1 Demonstrate an understanding of the properties (e.g. sides, angles, and diagonals) of squares and rectangles** E.g. 1 Give learners photocopied worksheets with shapes to identify squares and rectangles and give reasons. (Or Draw on the chalk board)  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning Look for Patterns and Relationships

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<tr>
<td>B5.3.1.1

Identify and sort quadrilaterals, including rectangles, squares, trapezoids, parallelograms, and rhombuses, according to their attributes CONT'D</td>
<td>B5.3.1.1.2 Use paper folding with cut-out squares and rectangles to investigate sides, angles, and diagonals properties

E.g. 1 Give learners foldable cut-out 2D squares and rectangles to investigate the properties in terms of sides, angles, and diagonals and complete the table

<table>
<tr><th>Properties</th><th>Rectangle</th><th>Square</th></tr>
<tr><td>Sides</td><td></td><td></td></tr>
<tr><td>All sides are congruent</td><td>No</td><td>Yes</td></tr>
<tr><td>Opposite sides are congruent</td><td></td><td></td></tr>
<tr><td>Opposite sides are parallel</td><td></td><td></td></tr>
<tr><td>Angle</td><td></td><td></td></tr>
<tr><td>Opposite angles are congruent</td><td></td><td></td></tr>
<tr><td>All angles are right angles</td><td></td><td></td></tr>
<tr><td>Diagonals</td><td></td><td></td></tr>
<tr><td>Diagonals bisect each other</td><td></td><td></td></tr>
<tr><td>Diagonals are congruent</td><td></td><td></td></tr>
<tr><td>Diagonals meet at right angles</td><td></td><td></td></tr>
</table>
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<tr><td>B5.3.1.1

Identify and sort quadrilaterals, including rectangles, squares, trapezoids, parallelograms, and rhombuses, according to their attributes CONT'D</td><td>
B5.3.1.1.3 Prove that a polygon is regular by measuring the sides and angles or by folding and superimposing

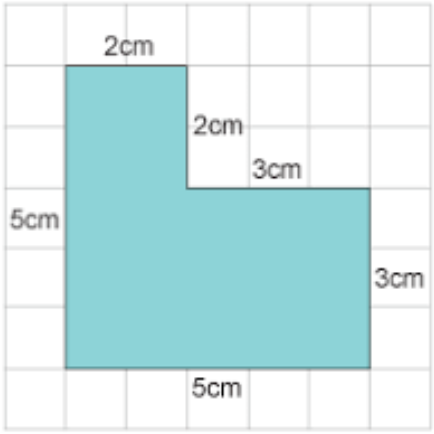
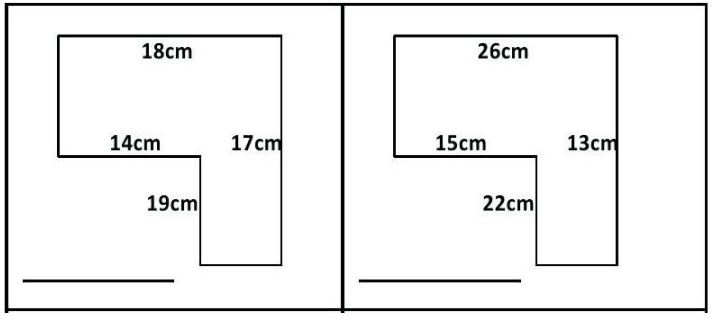
E.g. 1 Give learners foldable cut-out 2D quadrilaterals - rectangles, squares, trapezoids, parallelograms, and rhombuses - to investigate the properties in terms of sides, angles, and diagonals and complete the table
<table>
<tr><td>Properties</td><td>Rectangle</td><td>Square</td><td>rhombuses</td><td>trapezoids</td><td>parallelograms</td></tr>
<tr><td>Sides</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>All sides are congruent</td><td>No</td><td>Yes</td><td></td><td></td><td></td></tr>
<tr><td>Opposite sides are congruent</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Opposite sides are parallel</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Angle</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Opposite angles are congruent</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>All angles are right angles</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Diagonals</td><td></td><td></td><td></td><td></td><td></td></tr>
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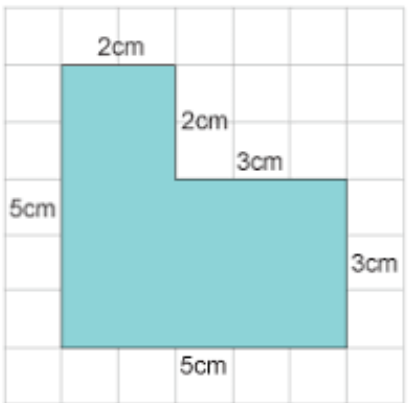
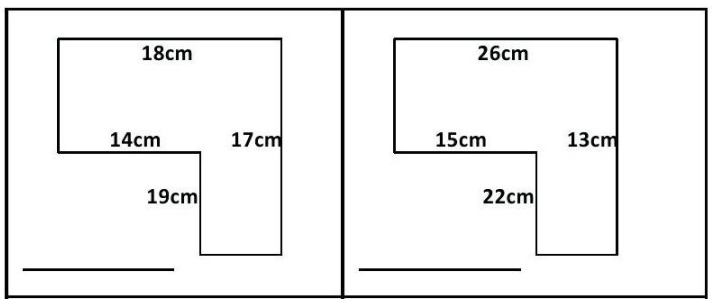
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.1 **Estimate and measure perimeter and surface area of 2-D shapes using centimetre and metre**	**B5.3.2.1.1 Estimate perimeter using referents for centimetre and metre, and calculate the actual perimeter and compare.** E.g. 1 Using pupils' referents for centimetre (hand span and thumb width) or metre (two strides or pace steps) estimate and calculate perimeter of any object in the classroom (e.g. exercise book, floor tiles, math set, teacher's table, classroom floor, etc.) then measure with a ruler of tape and calculate the actual perimeter and compare the answers. 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

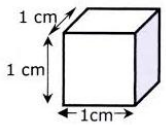
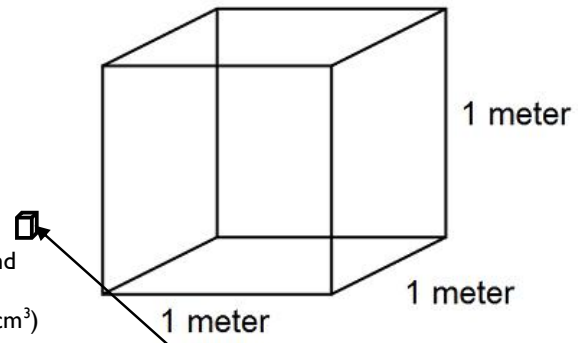
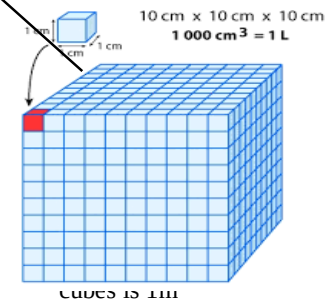
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.1 **Estimate and measure perimeter and surface area of 2-D shapes using centimetre and metre CONT'D**	**B5.3.2.1.2 Calculate perimeter of given shapes in centimetres and metres.** E.g. 1 Give learners 2D shapes with given dimensions to find the perimeter by adding the distance around the shapes   	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

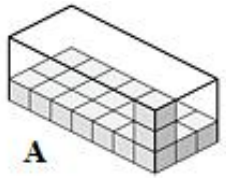
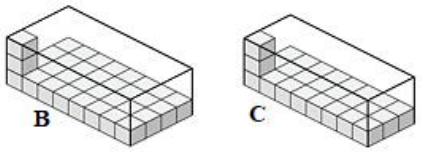
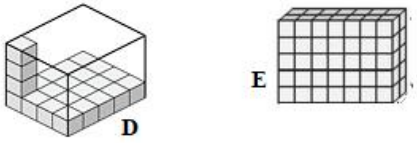
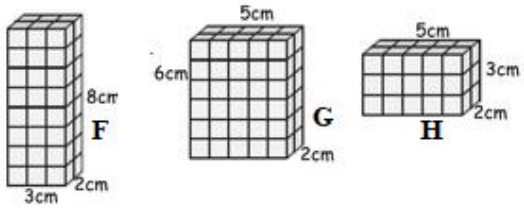
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.1 **Estimate and measure perimeter and surface area of 2-D shapes using centimetre and metre CONT'D**	**B5.3.2.1.3 Calculate surface area of given shapes in centimetres and metres.** E.g. 1 Give learners compound 2D shapes that can be spilt to many rectangles with given dimensions to find the surface area by adding areas of the various rectangles   	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership, Attention to Precision

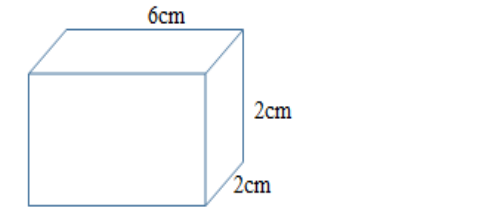
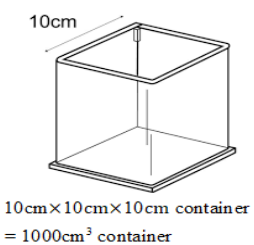
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.2 **Demonstrate an understanding of volume of common 3D shapes**	**B5.3.2.2.1 Select and justify referents for cm^3 or m^3 units.** E.g. 1 Show learners an object which is a **cubic centimetre (or 1 cm^3) and** another object which is a **cubic metre (or 1 m^3)**  Volume = 1 cm^3  E.g. 2 Let learners explore the classroom and compound to find that can be used as referents for the cubic centimetre ($1cm^3$) cubic metre ($1m^3$). Example of $1m^3$is polytank; and $1cm^3$is cube sugar. and E.g. 3 Ask learners identify containers in the classroom and compound that can be measured using cubic centimetre and cubic metre (m^3) (cm^3)  E.g. 4 Ask learners to explore and state the relationship between and $1m^3$; that is, a cubic is the volume of a cube with measurements 1cm×1cm×1cm, and is equal to1/1,000,000of a cubic meter,1/1,000 of a litre (**i.e.** $1m^3$=100cm×100cm×10cm =1,000,000cm^3). $1cm^3$ cubes is 1m	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

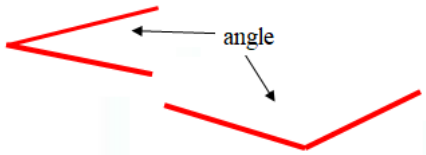
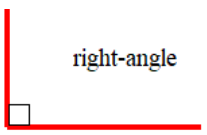
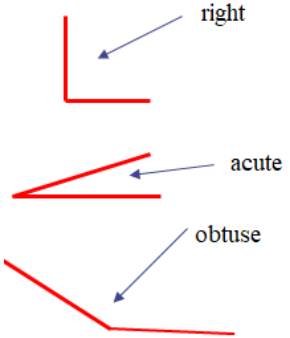
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.2 **Demonstrate an understanding of volume of common 3D shapes CONT'D**	**B5.3.3.2.2 Determine the volume of boxes by finding how many cubes of sizes 1cm³ each contains** E.g. 1 Ask learners to determine the volume of different sizes of boxes by investigating how many cubes of sizes 1cm³ can fill the containers. i.e. for box A, the base can take 2 by 7 1cm³, and to fill it this arrangement will done 3 times, hence Volume of A is base area × height = 2×7×3=42cm³  E.g. 2 Ask learners to determine the volume of the boxes B, C, D and E   E.g. 3 Guide learners to deduce the rule for volume of a box, and use is to calculate volumes of given boxes – F, G and H. That is, if the base area is $l \times w$ and the height is h, then the. Volume of box is given by $l \times w \times h$  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

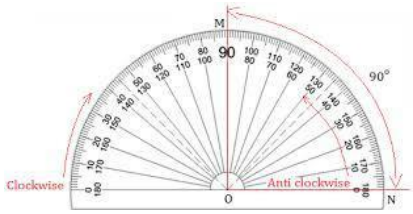
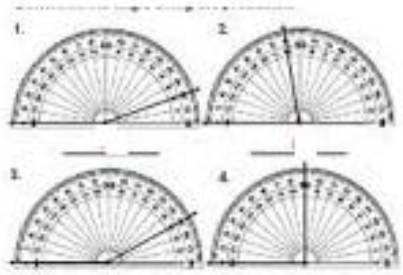
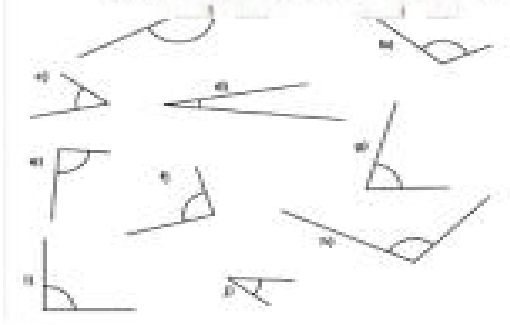
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.2.2 **Demonstrate an understanding of volume of common 3D shapes. CONT'D**	**B5.3.2.2.3Determine different sizes of boxes that have the same volume.** E.g. 1 Given 12 centimetre cubes, ask learners to find the different dimensions of boxes that can take the cubes. For 12 cubes the boxes can be of dimension 1cm×1cm× 12cm or 2cm×2cm×6cm. Can we find any more drawings. E.g. 2 Ask learners to investigate with 20, 24, 30 and 48 centimetre cubes and draw all possible boxes 	Learners develop: Problem solving skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision
	B5.3.3.2.4 Demonstrate an understanding of capacity by describing the relationship between capacity and volume as well as litre and 10cm³ E.g. 1 Give learners a litre container and a 10cm×10cm×10cm container (i.e. a 1000cm³container. Ask them to fill one with water and empty it into the other; and ask them to explore and state the relationship between 1cm³ and 1m³i.e. 10cm×10cm×10cm =1000cm³ = 1*litre*. [see B5.3.2.2.1 i.e. 1000litres is 1m³] Note: the volume is the actual amount of something which covers a definite space. Capacity is the potential amount of a substance which an object is able to hold (units are litres and millilitres , gallons, etc).  	

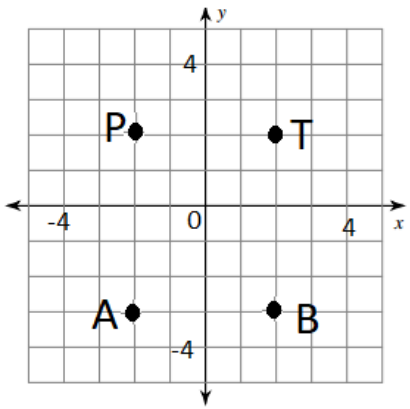
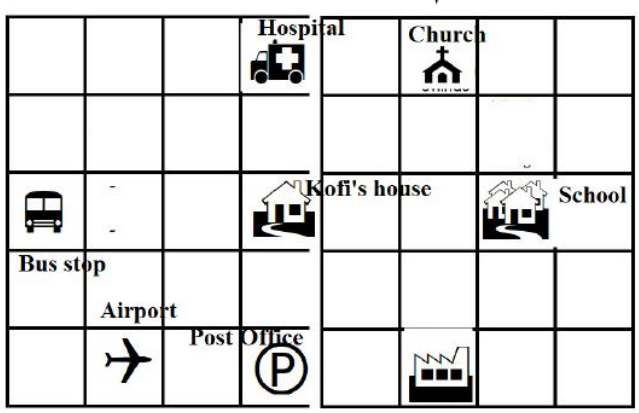
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.3.3 **Demonstrate an understanding of angles**	**B5.3.3.3.1 Identify examples of angles in the environment that are equal to, less than or larger than, a right angle** E.g. 1 Review the definition of angles: angles as the region formed in a plane when two lines meet or cross, they form an Angle. And ask learners to identified angles in the classroom and compound. E.g. opening laptop, door, window, pair of scissors, pavement floor, etc.  E.g. 2 Guide learners to see right angles and fold right angles from paper. Ask learners to identify examples of angles in the classroom and compound that are equal to, less than or larger than, a right angle       	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships

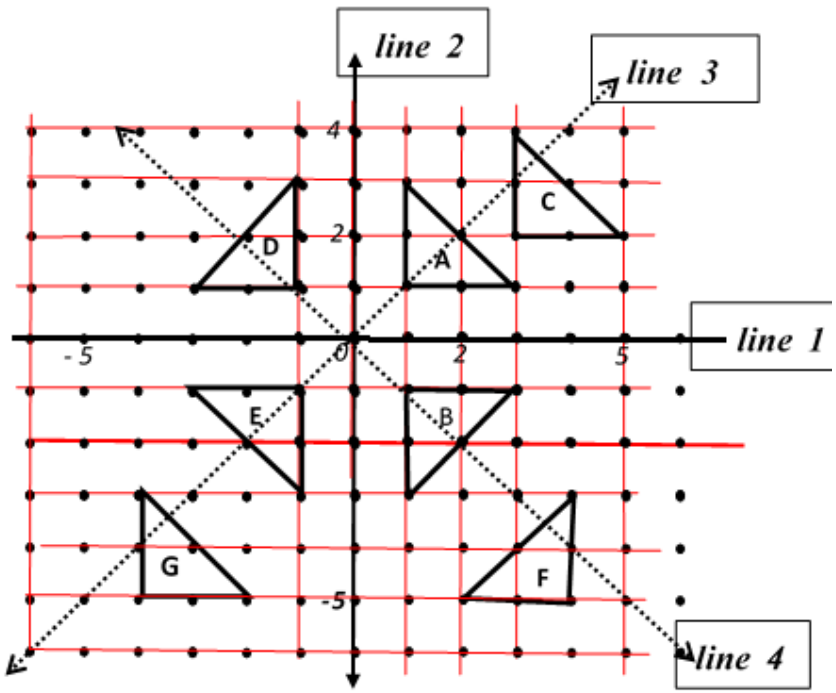
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.3.3 **Demonstrate an understanding of angles CONT'D**	**B5.3.3.3.2 Measure given angles with a protractor and classify them according to their measured sizes – right, acute, obtuse and reflex** E.g. 1 Review with learners how to use the protractor to measure angles  E.g. 2 Give learners photocopied worksheets with protractor in position measuring angles; ask them to read the angles or place their protractors on it and read it  E.g. 4 Give learners photocopied worksheets with several angles to measure and sort into those which are – right, acute obtuse angles. E.g. 5 Ask learners to draw various angles such as 30°, 45°,60°, 75°, 90°, 120°, 150°,270°, etc. 	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.4.1 **Describe the position of objects in space using the cardinal points**	**B5.3.4.1.1 Tell the position and motion of objects in space using the cardinal points north, south, east and west** E.g. 1. Pupils describe the position and motion of objects from one point to another using the 4 main cardinal points: e.g. The point **P** is to the west of point **T** but to the north of the **A** E.g. 2. Pupils describe the position of places in a grid from a given location. E.g. from Kofi's house - the school is East of Kofi's house; the hospital is North of Kofi's house E.g. 3. Ask learners to give directions from Kofi's house the various locations. E.g. Move 2 squares to the east and 2 squares to the north to get to the church  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.3.4.2 **Perform a single transformation (i.e. reflection) on a 2D shape**	**B5.3.4.2.2 Identify images of a single transformation (i.e. reflection) of a 2D shapes in a plane** E.g. 1. Ask learners to identify images of reflections of given 2D shapes in a given plane. Ask learners to study the shapes and the mirror line and state which are objects and their images in the given four mirror lines. Shape E is the image of Shape B in mirror line 2 or the line $x = 0$; shape A is a reflection of shape E in the line $y = x$.  	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Attention to Precision; Look for Patterns and Relationships

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<tr><td>B5.4.1.1

Differentiate between first-hand and second-hand data</td><td>
B5.4.1.1.1Explain the difference between first-hand and second-hand data.

E.g. 1. Explain first-hand data as data that you collect on your own; and second-hand data is data that has been collected and organized by somebody else.

E.g. 2. Mrs. Acquaye's class studied the weather. The learners measured the rainfall for 5 days (Monday to Friday). For her class these results are first-hand data. But for another class these results are second-hand data.

B5.4.1.1.2 Find examples of second-hand data in print and electronic media, such as newspapers, magazines, and the internet.

E.g. 1. Look for examples of second-hand data in the newspaper or website; You can find data on league table, price list, sales, rainfall, etc.

E.g. 2. Ask learners to study the second-hand data on the league table and write questions for their friends to read and interpret the data. E.g. Which team has not lost in a match? Which team has won only one match?

<table>
<tr><th>Team</th><th>P</th><th>W</th><th>D</th><th>L</th><th>F</th><th>A</th><th>Pts</th></tr>
<tr><td>Hearts</td><td>2</td><td>2</td><td>0</td><td>0</td><td>4</td><td>1</td><td>6</td></tr>
<tr><td>Kotoko</td><td>2</td><td>1</td><td>0</td><td>1</td><td>3</td><td>3</td><td>3</td></tr>
<tr><td>Dwafts</td><td>2</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>Gold Feilds</td><td>2</td><td>0</td><td>1</td><td>1</td><td>0</td><td>2</td><td>1</td></tr>
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</td><td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
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<tr><td>B5.4.1.1

Differentiate between first-hand and second-hand data. CONT'D</td><td>E.g. 3. Ask learners to study the second-hand data on suggested number of 30 minutes periods that Ghanaian primary school children learn certain subjects every week. Ask them to write questions for their friends to read and interpret the data.

<table>
<tr><th colspan="4">Number of weekly periods in each class</th></tr>
<tr><th rowspan="2">Subject</th><th colspan="3">Upper primary</th></tr>
<tr><th>P4</th><th>P5</th><th>P6</th></tr>
<tr><td>English</td><td>10</td><td>10</td><td>10</td></tr>
<tr><td>Mathematics</td><td>10</td><td>10</td><td>10</td></tr>
<tr><td>National language</td><td>6</td><td>6</td><td>6</td></tr>
<tr><td>Environmental studies</td><td>5</td><td>5</td><td>5</td></tr>
<tr><td>Integrated sciences</td><td>5</td><td>5</td><td>5</td></tr>
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Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
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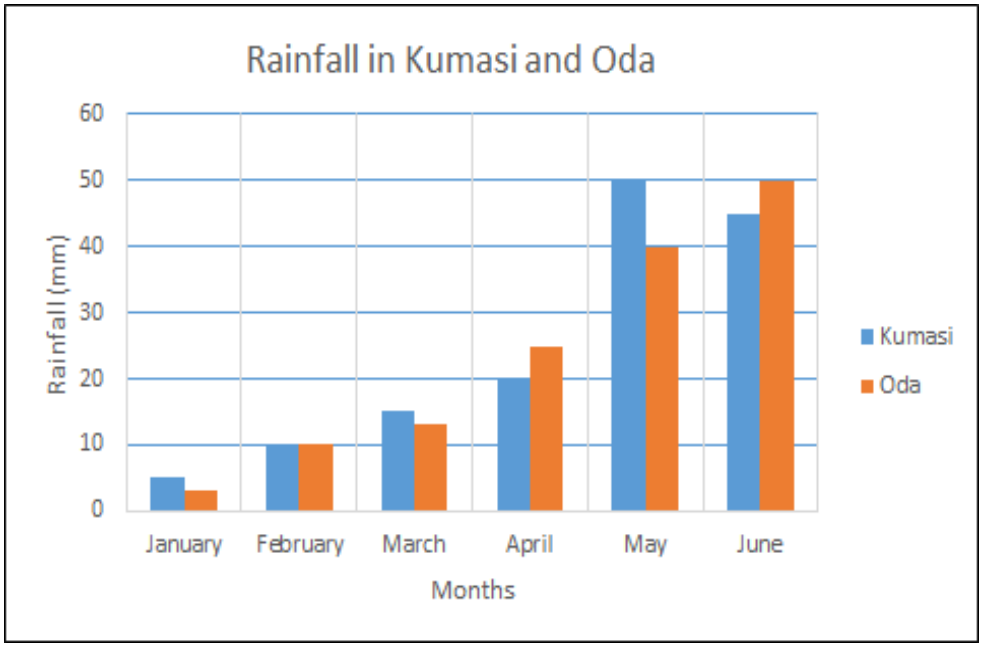
Construct and interpret double bar graphs to draw conclusions</td><td>B5.4.1.2.1. Draw double bar graphs, complete with title, labelled axes, key or legend, to represent data collected (up to 3 pairs of categories of data and use it to solve problems.

E.g. 1. Give students data presented in a table on rainfall in mm for two towns to draw a double bar graph complete with title, labelled axes, key

<table>
<tr><th></th><th>January</th><th>February</th><th>March</th><th>April</th><th>May</th><th>June</th></tr>
<tr><td>Kumasi</td><td>5</td><td>10</td><td>15</td><td>20</td><td>50</td><td>45</td></tr>
<tr><td>Oda</td><td>3</td><td>10</td><td>13</td><td>25</td><td>40</td><td>50</td></tr>
</table></td><td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning Look for Patterns and Relationships</td></tr>
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.4.1.2 **Construct and interpret double bar graphs to draw conclusions** **CONT'D**		Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision

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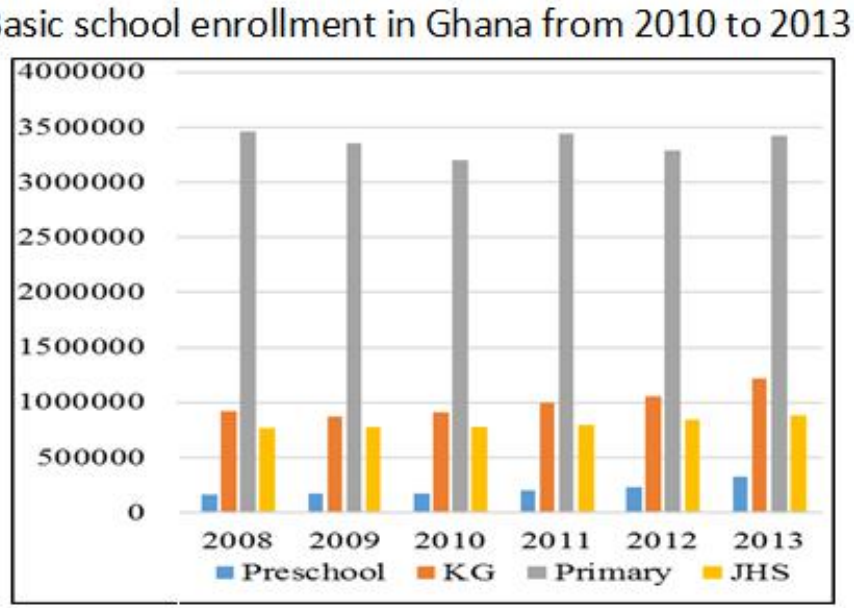
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<tr><td>B5.4.1.2
Construct and interpret double bar graphs to draw conclusions
CONT'D</td><td>E.g. 2. Give students more paired data presented in a tables. Ask them to study the paired data in the tables and ask them questions based on them. Also ask them to write questions for their friends to read and interpret the data. For instance some questions can be (i) In which subject was Fusena's worse performance? (ii) How many pupils are in the upper primary classes in Presby? (iii) When was the best day for Senasco shop?

E.g. 3. Ask learners to draw double bar graphs complete with title, labelled axes, key for the paired data presented in the tables on (i) percentage test scores of 2 students; (ii) enrolment in 2 schools in a town; sales of two of bread in two shops in a week; etc.

(i) test scores of 2 students
<table><tr><td>Student</td><td>Maths</td><td>English</td><td>Ghanaian Language</td><td>Science</td><td>Social Studies</td></tr><tr><td>Fusena</td><td>75</td><td>60</td><td>80</td><td>60</td><td>65</td></tr><tr><td>Alidu</td><td>55</td><td>70</td><td>75</td><td>70</td><td>55</td></tr></table>
(ii) enrollment in 2 school in a town
<table><tr><td>School</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td><td>P6</td></tr><tr><td>Presby</td><td>35</td><td>30</td><td>40</td><td>30</td><td>35</td><td>45</td></tr><tr><td>Roman</td><td>45</td><td>50</td><td>55</td><td>50</td><td>65</td><td>60</td></tr></table>
(iii) Sales of bread in two shops in a week
<table><tr><td>Shop</td><td>Monday</td><td>Tuesday</td><td>Wednesday</td><td>Thursday</td><td>Friday</td></tr><tr><td>Abijak</td><td>25</td><td>22</td><td>20</td><td>24</td><td>24</td></tr><tr><td>Senasco</td><td>25</td><td>18</td><td>16</td><td>22</td><td>26</td></tr></table></td><td>Learners develop:

Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning
Look for Patterns and Relationships</td></tr>
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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.4.1.2 Construct and interpret double bar graphs to draw conclusions CONT'D	 **B5.4.1.2.2 Identify examples of double bar graphs used in a variety of print and electronic media, such as newspapers, magazines, and the internet.** E.g. 1. Provide opportunities for learners to access, read and interpret examples of double bar graphs used in a variety of print and electronic media, such as newspapers, magazines and the internet	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership; Attention to Precision

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CONTENT STANDARDS	INDICATORS AND EXEMPLARS	SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES
B5.4.2.1 **Describe the likelihood of a single outcome occurring using words such as impossible, possible, and certain**	**B5.4.2.1.1 Classify the likelihood of a single outcome occurring in a probability experiment as impossible, possible, or certain** E.g. In a Ludo competition with a die, the event of "obtaining the number 1" is possible; the event of "obtaining the number 7" is impossible; the event of "obtaining the number 1 or 2 or 3 or 4 or 5 or 6" is certain.  **B5.4.2.1.2 Design and conduct an experiment in which the likelihood of a single outcome occurring is impossible, possible (likely or unlikely), certain.** E.g. In a bag containing 10 red, 4 green and 1 pink bottle tops, let a learner pick one bottle top from the bag. 1. Picking a black bottle top is impossible 2. Picking a red bottle top is possible (likely) 3. Picking a pink bottle top is possible (unlikely) 4. Picking a red or green or pink bottle top is certain. Explain each of the four outcomes **B5.4.2.1.3 Conduct a given probability experiment a number of times, recording the outcomes, and explaining the results.** E.g. Ask 5 learners of their expectations/predictions about the outcome of tossing a coin three times. Teacher tosses a coin three times and records the outcome. (e.g. head, head, tail) Does the outcome match their expectation? Either way the teacher should explain. In pairs, learners should repeat the experiment several times, record the outcomes and compare the number of heads turning up	Learners develop: Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership; Attention to Precision

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